

Linear Algebra Comprehensive Review Sheet

Complete Topic Summaries, Master Theorems, and Computational Algorithms

Grounded in David C. Lay's *Linear Algebra and Its Applications* (Global Edition)

1. Vector Spaces, Subspaces, and Row Reduction

1.1 Vector Spaces and Subspaces (§4.1)

- **Vector Space (V):** A nonempty set of objects with vector addition and scalar multiplication satisfying the 10 formal axioms (closure, associativity, commutativity, zero vector $\mathbf{0}$, additive inverses, and distributivity).
- **Subspace Criteria (Theorem 1):** A subset $H \subseteq V$ is a subspace if and only if:
 - (i) The zero vector $\mathbf{0} \in H$.
 - (ii) Closed under vector addition: $\mathbf{u}, \mathbf{v} \in H \implies \mathbf{u} + \mathbf{v} \in H$.
 - (iii) Closed under scalar multiplication: $\mathbf{u} \in H, c \in \mathbb{R} \implies c\mathbf{u} \in H$.
- **Spanning Theorem for Subspaces:** If $\mathbf{v}_1, \dots, \mathbf{v}_p \in V$, then $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is guaranteed to be a subspace of V .

1.2 Row Reduction and Echelon Forms (§1.2)

- **Echelon Form (REF):** (1) All nonzero rows above any zero rows; (2) each leading entry is strictly to the right of the leading entry above it; (3) all entries below a leading entry are zero.
- **Reduced Echelon Form (RREF):** Echelon form where every leading entry is 1, and each leading 1 is the *sole* nonzero entry in its column.
- **Variables:** Columns with pivots correspond to **basic variables**; columns without pivots correspond to **free variables**.
- **System Solvability:** A linear system is consistent $\iff$ the echelon form of $[A \mid \mathbf{b}]$ contains no row $[0 \ 0 \ \dots \ 0 \mid b]$ with $b \neq 0$. If consistent, the solution is unique when there are no free variables, and has infinitely many solutions if at least one free variable exists.

2. Matrix Inverses and Factorizations

2.1 Matrix Inversion (§2.2)

- A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible (nonsingular) $\iff \exists A^{-1}$ such that $AA^{-1} = A^{-1}A = I_n$.
- **2×2 Inversion Formula:** For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, if $\det(A) = ad - bc \neq 0$:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- **Inversion Algorithm (Theorem 7):** Row reduce the augmented matrix:

$$[A \mid I_n] \sim [I_n \mid A^{-1}]$$

If A cannot be reduced to I_n , then A is singular (not invertible).

- **Linear Systems:** If A is invertible, $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.
- **Independence of Columns:** If A is invertible, $A\mathbf{x} = \mathbf{0} \implies \mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}$, meaning the columns of A are linearly independent.

2.2 LU Factorization Algorithm (§2.5)

Algorithm: Solving Linear Systems via LU Factorization

Structure: $A = LU$, where L is a unit lower triangular matrix (1's on main diagonal, 0's above) and U is an upper triangular echelon form of A .

1. **Forward Substitution:** Solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ from top to bottom.
2. **Back Substitution:** Solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ from bottom to top.

Constructing L and U :

- Reduce A to echelon form U using *row replacements only* ($R_i \rightarrow R_i - cR_j$ with $i > j$). No row interchanges.
- In each column where a pivot is chosen, divide the entries on and below the pivot by the pivot value, and place them into the corresponding column of L .

3. Fundamental Subspaces, Bases, and Dimension

3.1 Null Spaces, Column Spaces, and Linear Transformations (§4.2)

Property	Null Space $\text{Nul } A$	Column Space $\text{Col } A$
Definition	$\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$ (homogeneous solutions)	$\{\mathbf{b} \in \mathbb{R}^m : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}$
Subspace of	$\mathbb{R}^n$ (n = number of columns of A)	$\mathbb{R}^m$ (m = number of rows of A)
Test Membership	Compute $A\mathbf{w}$; check if $A\mathbf{w} = \mathbf{0}$.	Solve $A\mathbf{x} = \mathbf{b}$; check consistency.
Finding a Basis	Solve $A\mathbf{x} = \mathbf{0}$ to RREF; write in parametric vector form. Free variable vectors form a basis.	Row reduce A to echelon form to find pivot columns. Select the original columns of A at those positions.

- **Linear Transformation** $T : V \rightarrow W$: $\ker(T) = \{\mathbf{u} \in V : T(\mathbf{u}) = \mathbf{0}\}$ is a subspace of V ; $\text{range}(T)$ is a subspace of W .

3.2 Linearly Independent Sets and Bases (§4.3)

- **Linear Independence**: $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ holds **only** when $c_1 = \cdots = c_p = 0$.
- **Basis of Subspace** H : A set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ that is: (1) linearly independent, and (2) spans H .
- **Spanning Set Theorem**: If one vector in a spanning set is a linear combination of the others, it can be removed without altering the span.
- **Basis for Col A** : The pivot columns of A form a basis for $\text{Col } A$.

3.3 Dimension and the Rank Theorem (§4.5)

- $\dim V$ = number of vectors in any basis for V . By definition, $\dim\{\mathbf{0}\} = 0$.
- $\text{rank } A = \dim(\text{Col } A) = \dim(\text{Row } A)$ = number of pivot positions in A .
- $\text{nullity } A = \dim(\text{Nul } A)$ = number of free variables in $A\mathbf{x} = \mathbf{0}$.
- **The Rank-Nullity Theorem (Master Relationship)**: For any $m \times n$ matrix A :

$$\text{rank } \mathbf{A} + \text{nullity } \mathbf{A} = \mathbf{n} \quad (\text{total number of columns})$$

4. Coordinate Systems and Change of Basis

4.1 Coordinate Systems (§4.4)

- **Unique Representation Theorem**: If $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for V , each $\mathbf{x} \in V$ can be uniquely written as:

$$\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n \iff [\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

- **Change-of-Coordinates Matrix to Standard Basis**: $P_{\mathcal{B}} = [\mathbf{b}_1 \ \dots \ \mathbf{b}_n]$. Then:

$$\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}} \quad \text{and} \quad [\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$$

- **Coordinate Isomorphism**: The mapping $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a one-to-one and onto linear transformation from V to $\mathbb{R}^n$ ($V \cong \mathbb{R}^n$).

4.2 Change of Basis (§4.6)

Change-of-Coordinates Matrix Between Arbitrary Bases

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ be bases for a vector space V .

- **Coordinate Conversion**: $[\mathbf{x}]_{\mathcal{C}} = \underset{\mathcal{C} \leftarrow \mathcal{B}}{P} [\mathbf{x}]_{\mathcal{B}}$.
- **Matrix Definition**: The columns of $\underset{\mathcal{C} \leftarrow \mathcal{B}}{P}$ are the $\mathcal{C}$ -coordinate vectors of the basis $\mathcal{B}$:

$$\underset{\mathcal{C} \leftarrow \mathcal{B}}{P} = [\mathbf{[b_1]_C} \ \mathbf{[b_2]_C} \ \dots \ \mathbf{[b_n]_C}]$$

- **Row Reduction Algorithm in $\mathbb{R}^n$** :

$$[\mathbf{c_1} \ \dots \ \mathbf{c_n} \mid \mathbf{b_1} \ \dots \ \mathbf{b_n}] \sim \left[\begin{array}{c|c} I_n & \underset{\mathcal{C} \leftarrow \mathcal{B}}{P} \end{array} \right]$$

- **Inversion Property**: $\underset{\mathcal{B} \leftarrow \mathcal{C}}{P} = \left(\underset{\mathcal{C} \leftarrow \mathcal{B}}{P} \right)^{-1}$.

5. Orthogonality, Projections, and the Gram–Schmidt Process

5.1 Inner Product, Length, and Distance (§6.1)

- **Inner Product (Dot Product):** $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + \cdots + u_n v_n$.
- **Length (Norm):** $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}$. Furthermore, $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$.
- **Distance:** $\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
- **Orthogonality:** $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$. The zero vector $\mathbf{0}$ is orthogonal to every vector.
- **Pythagorean Theorem:** $\mathbf{u} \perp \mathbf{v} \iff \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

5.2 Orthogonal Sets and Projections onto a Line (§6.2)

- An orthogonal set of nonzero vectors is automatically **linearly independent** (Theorem 4).
- **Expansion in an Orthogonal Basis:** If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal basis for W , any $\mathbf{y} \in W$ satisfies:

$$\mathbf{y} = \sum_{j=1}^p c_j \mathbf{u}_j \quad \text{where} \quad c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}$$

- **Projection onto a Line $L = \text{Span}\{\mathbf{u}\}$:**

$$\hat{\mathbf{y}} = \text{proj}_L \mathbf{y} = \left(\frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \right) \mathbf{u}, \quad \text{Distance from } \mathbf{y} \text{ to } L = \|\mathbf{y} - \hat{\mathbf{y}}\|$$

- **Orthonormal Set:** An orthogonal set where every vector has unit length ($\|\mathbf{u}_i\| = 1$). A square matrix U has orthonormal columns $\iff U^T U = I \iff U^{-1} = U^T$.

5.3 Orthogonal Decomposition and General Projections (§6.3)

Orthogonal Decomposition Theorem & Best Approximation

Let W be a subspace of $\mathbb{R}^n$ with **orthogonal basis** $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$.

1. **Orthogonal Projection of $\mathbf{y}$ onto W :**

$$\hat{\mathbf{y}} = \text{proj}_W \mathbf{y} = \left(\frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \right) \mathbf{u}_1 + \left(\frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \right) \mathbf{u}_2 + \cdots + \left(\frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \right) \mathbf{u}_p$$

2. **Orthogonal Component:** $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} \in W^\perp$, giving the unique decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$.
3. **Best Approximation Theorem:** $\hat{\mathbf{y}}$ is the uniquely closest point in W to $\mathbf{y}$:

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\| \quad \text{for all } \mathbf{v} \in W, \mathbf{v} \neq \hat{\mathbf{y}}$$

5.4 The Gram–Schmidt Process and QR Factorization (§6.4)

Gram–Schmidt Orthogonalization and QR Factorization

Given a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ for a subspace W , construct an orthogonal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ by:

$$\begin{aligned} \mathbf{v}_1 &= \mathbf{x}_1 \\ \mathbf{v}_2 &= \mathbf{x}_2 - \left(\frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1 \\ \mathbf{v}_3 &= \mathbf{x}_3 - \left(\frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1 - \left(\frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \right) \mathbf{v}_2 \\ &\vdots \\ \mathbf{v}_p &= \mathbf{x}_p - \sum_{j=1}^{p-1} \left(\frac{\mathbf{x}_p \cdot \mathbf{v}_j}{\mathbf{v}_j \cdot \mathbf{v}_j} \right) \mathbf{v}_j \end{aligned}$$

Orthonormal Basis: Normalize each vector: $\mathbf{u}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$.

QR Factorization: If $A \in \mathbb{R}^{m \times n}$ has linearly independent columns, then $A = QR$ where:

- Q is $m \times n$ with orthonormal columns ($Q^T Q = I_n$).
- $R = Q^T A$ is an $n \times n$ upper triangular invertible matrix with positive diagonal entries.
- $\text{Col } A = \text{Col } Q$.

6. Least-Squares Problems and Symmetric Diagonalization

6.1 Least-Squares Problems (§6.5)

Solving Inconsistent Systems: Least-Squares Problems

Goal: Find $\hat{\mathbf{x}} \in \mathbb{R}^n$ that minimizes the error $\|\mathbf{b} - A\mathbf{x}\|$.

1. **The Normal Equations (Theorem 13):** Every least-squares solution satisfies:

$$\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{b}$$

2. **Unique Solution Theorem (Theorem 14):** $A^T A$ is invertible $\iff$ columns of A are linearly independent. In this case:

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

3. **Column Space Projection:** $A\hat{\mathbf{x}} = \hat{\mathbf{b}} = \text{proj}_{\text{Col } A} \mathbf{b}$. The least-squares error is $\|\mathbf{b} - A\hat{\mathbf{x}}\|$.

6.2 Diagonalization of Symmetric Matrices (§7.1)

Orthogonal Diagonalization of Symmetric Matrices

Symmetric Matrix: A square matrix A is symmetric if $A^T = A$.

- **Fundamental Theorem (Theorem 2):** An $n \times n$ matrix A is orthogonally diagonalizable ($A = PDP^T$ with $P^{-1} = P^T$ and D diagonal) $\iff A$ is a symmetric matrix.
- **Eigenvector Orthogonality (Theorem 1):** Eigenvectors corresponding to *distinct* eigenvalues of a symmetric matrix are mutually orthogonal.
- **Full Step-by-Step Procedure:**
 1. Compute eigenvalues by solving characteristic equation $\det(A - \lambda I) = 0$.
 2. For each distinct eigenvalue λ , find a basis for its eigenspace $\text{Nul}(A - \lambda I)$.
 3. If an eigenspace has dimension > 1 , apply Gram–Schmidt to construct an orthogonal basis for that eigenspace.
 4. Normalize all eigenvectors to length 1.
 5. Construct $P = [\mathbf{u}_1 \ \dots \ \mathbf{u}_n]$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

7. Master Comparison Matrix and Exam Traps

7.1 Master Matrix of Major Decompositions

Factorization	Matrix Form	Conditions	Key Component Properties	Primary Application
LU	$A = LU$	No row swaps required	L unit lower triangular, U echelon upper triangular	Fast repeated solution of $A\mathbf{x} = \mathbf{b}$.
QR	$A = QR$	Linearly independent cols	Q has orthonormal cols ($Q^T Q = I_n$), R upper triangular	Stable least-squares: $R\hat{\mathbf{x}} = Q^T \mathbf{b}$.
Orthogonal Diag.	$A = PDP^T$	A is symmetric ($A^T = A$)	P orthogonal ($P^{-1} = P^T$), D real diagonal	Spectral decomposition, quadratic forms.

7.2 High-Yield Exam Checklist & Common Pitfalls

Frequent Pitfalls to Avoid in Linear Algebra Exams

- **Basis for Col A vs. RREF columns:** Row reduction changes the column space! Locate pivot columns using $\text{RREF}(A)$, but **always select the corresponding columns from the original matrix A** .
- **Basis for Row A :** Unlike columns, row operations do *not* change the row space. The nonzero rows of the echelon form **do** form a basis for Row A .
- **Ambient Spaces for Nul A and Col A :** For an $m \times n$ matrix A , remember $\text{Nul } A \subseteq \mathbb{R}^n$ (domain), whereas $\text{Col } A \subseteq \mathbb{R}^m$ (codomain).
- **When is $A^T A$ invertible?** $A^T A$ is invertible $\iff$ the columns of A are *linearly independent*. If columns are dependent, $A^T A \mathbf{x} = A^T \mathbf{b}$ is still consistent, but has infinitely many least-squares solutions.
- **Orthogonal Matrix Definition:** An “orthogonal matrix” is always **square** and has **orthonormal** columns ($P^T P = I \implies P^{-1} = P^T$). Always normalize eigenvectors when constructing P .